

Complex Disordered Systems

Gelation

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Today

- Physical gels and percolation
- Colloid-polymer mixtures
- Structure and characterization
- Arrested spinodal decomposition

What is a Physical Gel?

Physical gel: Particles/polymers connected via **reversible**, non-covalent bonds forming a **percolating network**

Key features:

- Bonds break/reform dynamically
- Energy $\sim k_B T$
- Self-healing possible
- **Thermoreversible**

Examples:

- Gelatin
- Agarose
- Yoghurt
- Paint, inks

Cross-links form via dense local regions (microcrystalline/glassy)

Physical vs Chemical Gels

Physical Gels

- Reversible bonds
- Energy $\sim k_B T$
- Dynamic network
- Thermoreversible

Key transition:

- Fluid $\rightarrow$ Gel occurs when a **system-spanning cluster** forms (**percolation**)
- Dramatic viscosity increase + elastic response

Gels are a second example of disordered solids (after glasses)

Chemical Gels

- Irreversible covalent bonds
- High energy barriers
- Permanent network

Percolation Theory

Problem: Square grid $L \times L$ sites, randomly label N of them. When does a spanning cluster form?

- $p = N/L^2$ = occupation probability
- **Percolation threshold** p_c
- Below p_c : small clusters
- Above p_c : giant spanning cluster

2D square lattice:

$$p_c \approx 0.5927$$

Transition: Continuous (2nd order)

Percolation Simulation

Python Code

↺ Start Over

▶ Run Code

```
1 p, L = 0.5, 120
2 grid, mask = sample_percolation(L,p)
3 plt.figure(figsize=(6, 6))
4 plt.imshow(grid, cmap='gray', interpolation='none')
5 plt.imshow(np.ma.masked_where(~mask, mask), cmap='autumn', alpha=0.8, interpolation='none')
6 plt.axis('off');plt.title(f'Percolation in 2D (p={p})\nLargest cluster highlighted in red');plt.show()
```

Percolation Transition

Python Code

↺ Start Over

▶ Run Code

```
1 plot_transition(npoints=50)
```

name 'sample_percolation' is not defined

Percolation Threshold: p_c Values

Non-universal: p_c depends on lattice type and dimension

Lattice	Dim	z	p_c^{site}	p_c^{bond}
Square	2	4	0.593	0.5
Triangular	2	6	0.5	0.347
Cubic	3	6	0.312	0.249
Hypercubic	4	8	0.197	0.160
Hypercubic	6	12	0.109	0.094

Higher dimensions $\rightarrow$ lower p_c (easier to percolate)

Critical Exponents: Universal

Universal: Critical exponents independent of lattice details

$$P_{\infty}(p) \propto (p - p_c)^{\beta}$$

Dimension d	β
2	$5/36 \approx 0.14$
3	≈ 0.41
≥ 6	1 (mean-field)

Gels require percolation for mechanical stability!

Colloidal Gel Formation

MD simulation of dilute colloidal gel formation (largest cluster in red)

Percolating cluster → mechanical rigidity

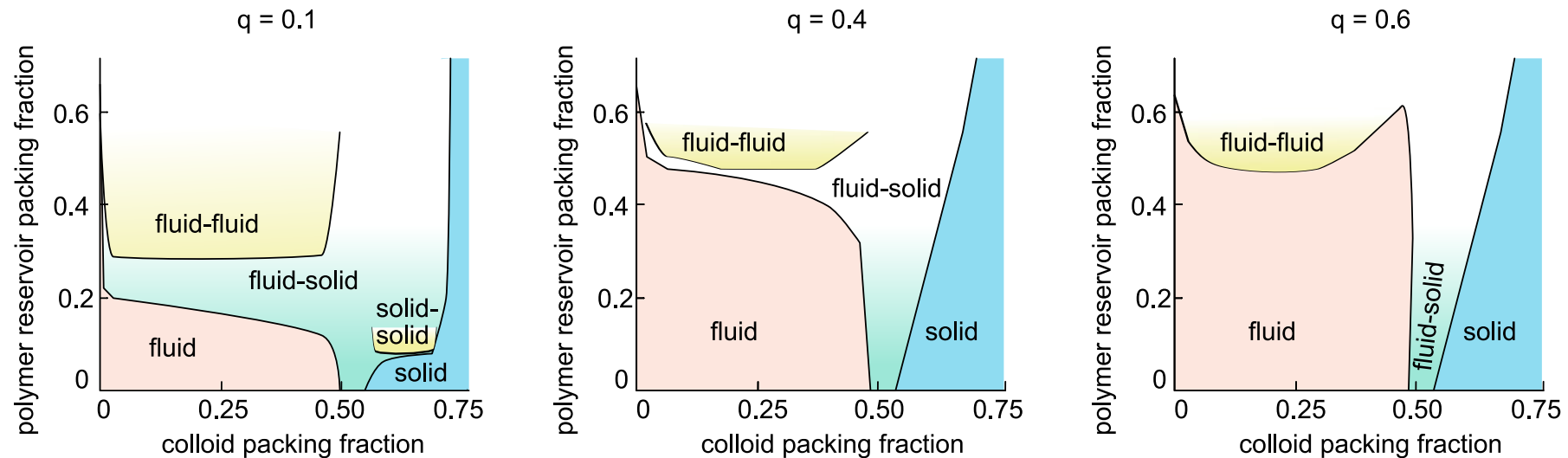
Colloid-Polymer Mixtures

Depletion interactions like the Asakura-Oosawa model produce a **very short ranged** potential with strong **attractive part**.

$$W_{AO}(r) = -\frac{4\pi\rho_p k_B T}{3} R^3 (1+q)^3 \left[1 - \frac{3}{4} \frac{r}{R(1+q)} + \frac{1}{16} \left(\frac{r}{R(1+q)} \right)^3 \right], \quad 2R < r < 2(R + R_p), \quad q = \frac{R_p}{R}$$

attractive + very short ranged $\rightarrow$ **sticky** interaction

- Colloids cluster into nonequilibrium branched phase
- Binodal becomes **metastable** to fluid-solid coexistence



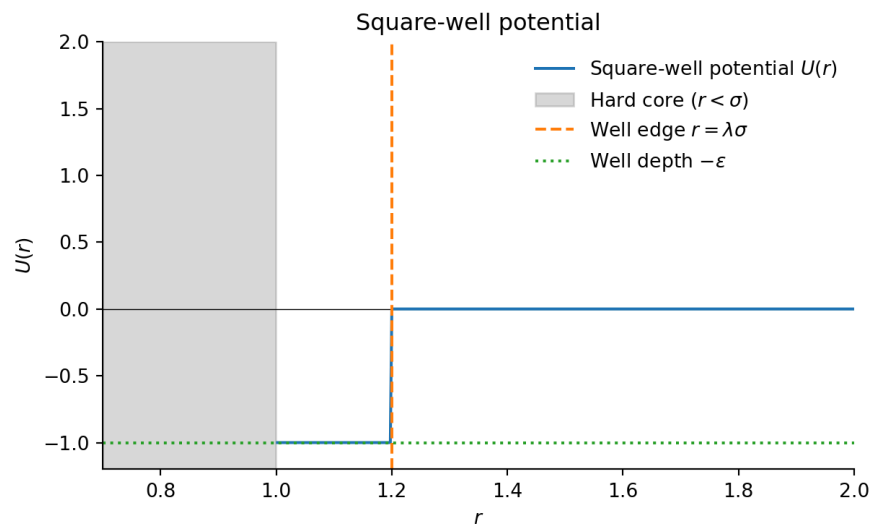
Phase diagram of colloid-polymer mixtures for varying polymer-colloid aspect ration q . Small

Simple Model Potentials

Square-well potential:

$$U(r) = \begin{cases} \infty & r < \sigma \\ -\epsilon & \sigma \leq r < \lambda\sigma \\ 0 & r \geq \lambda\sigma \end{cases}$$

- σ : particle diameter
- ϵ : well depth
- λ : range parameter



Morse Potential (MD Simulations)

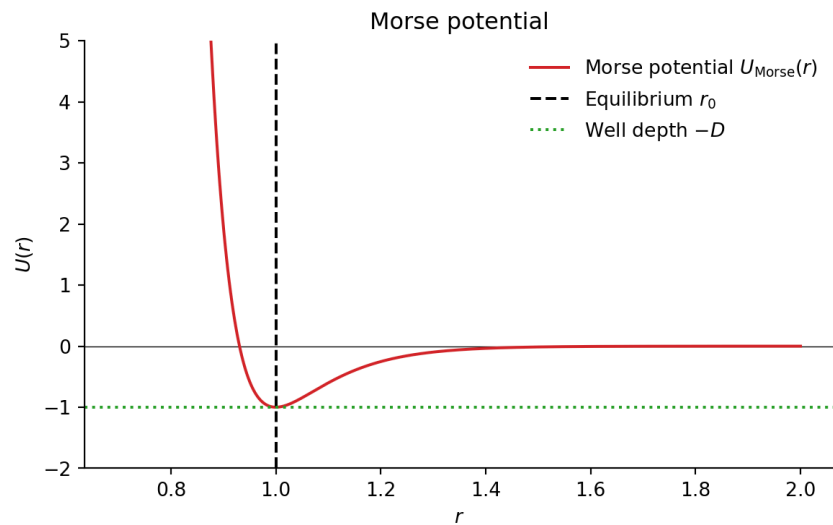
Smooth short-range attractive potential:

$$U_{\text{Morse}}(r) = D \left[e^{-2\alpha(r-r_0)} - 2e^{-\alpha(r-r_0)} \right]$$

- D : well depth
- α : steepness (controls range)
- r_0 : equilibrium distance



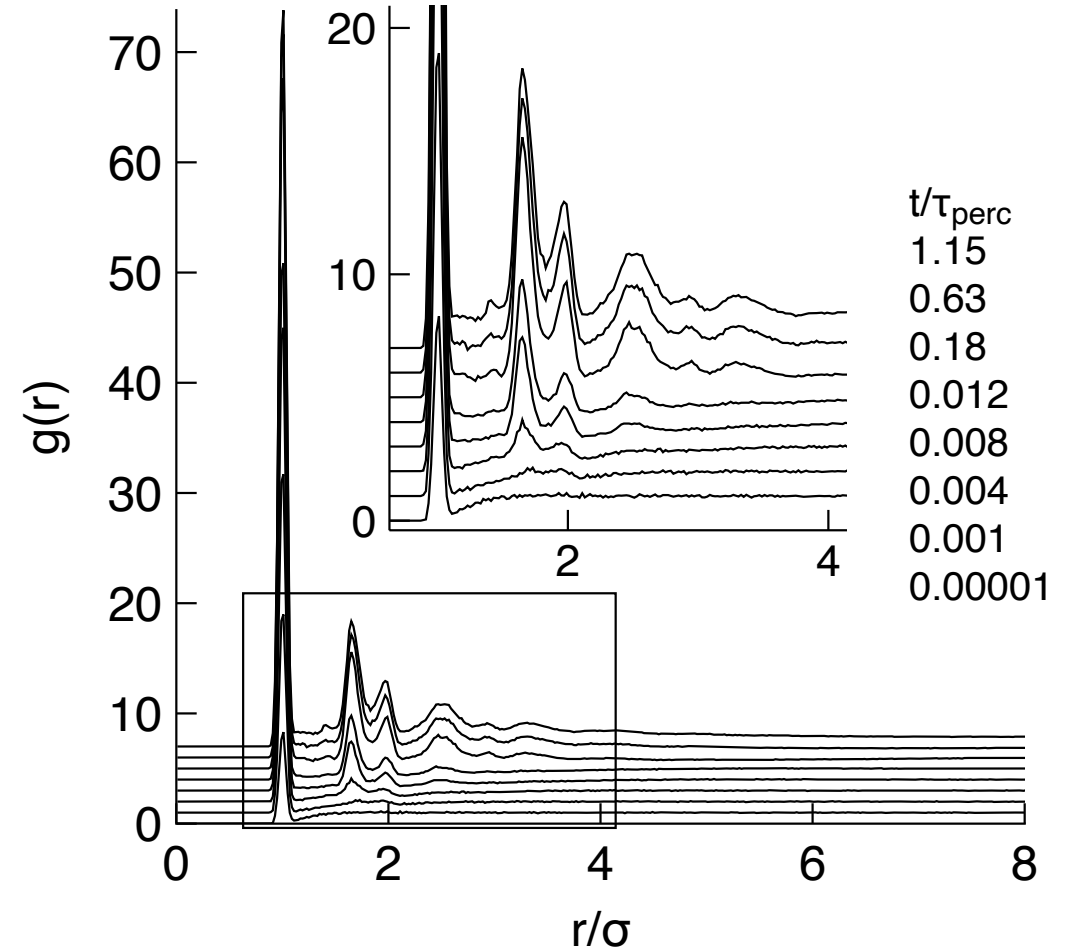
Morse potential placeholder



Pair Correlations: $g(r)$

Radial distribution function $g(r)$: best at short distances

- Captures cluster structure
- Sharp peaks at dilute concentrations
- Nearest/second-nearest neighbor shells



Time evolution of $g(r)$ in a dilute gel in units of the time to percolate τ_{perc}

Structure Factor: $S(q)$ and Fractals

Near percolation threshold, the log-q trend from the structure factor links to the compressibility: **power-law behavior**: Clusters become **fractal objects** with self-similar structure at all length scales.

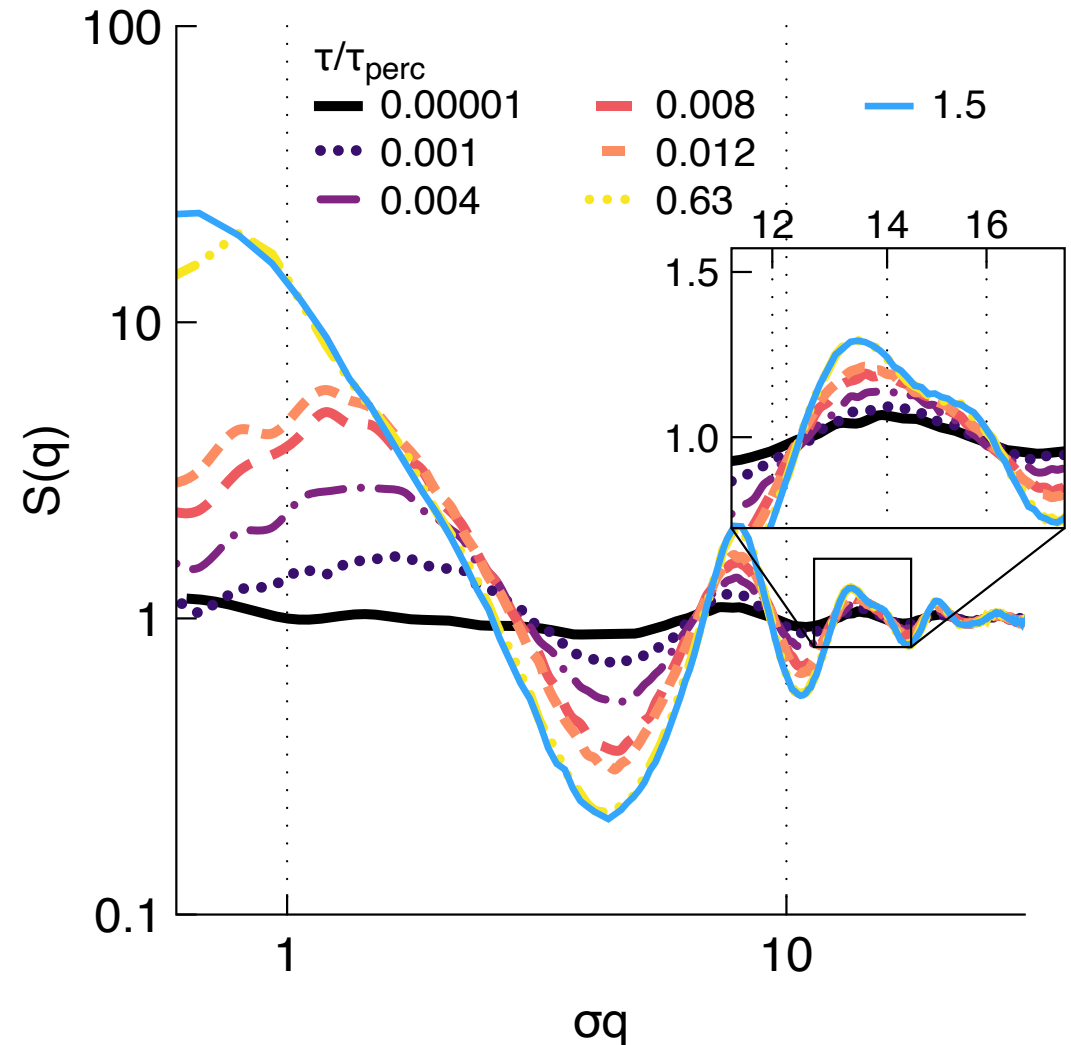
For a fractal object with dimension D_f :

- Mass scales as $M(R) \sim R^{D_f}$
- Density correlations decay as $\rho(r) \sim r^{-(d-D_f)}$

Fourier transform of power-law correlations gives:

$$S(q) \sim q^{-D_f}$$

- D_f : fractal dimension
- No characteristic length scale
- Typical 3D: $D_f \approx 2.5$



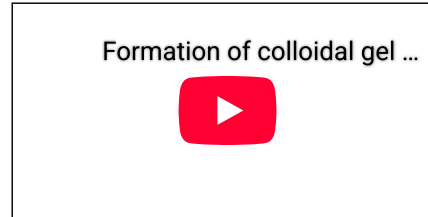
Arrested Spinodal Decomposition

How do gels form?

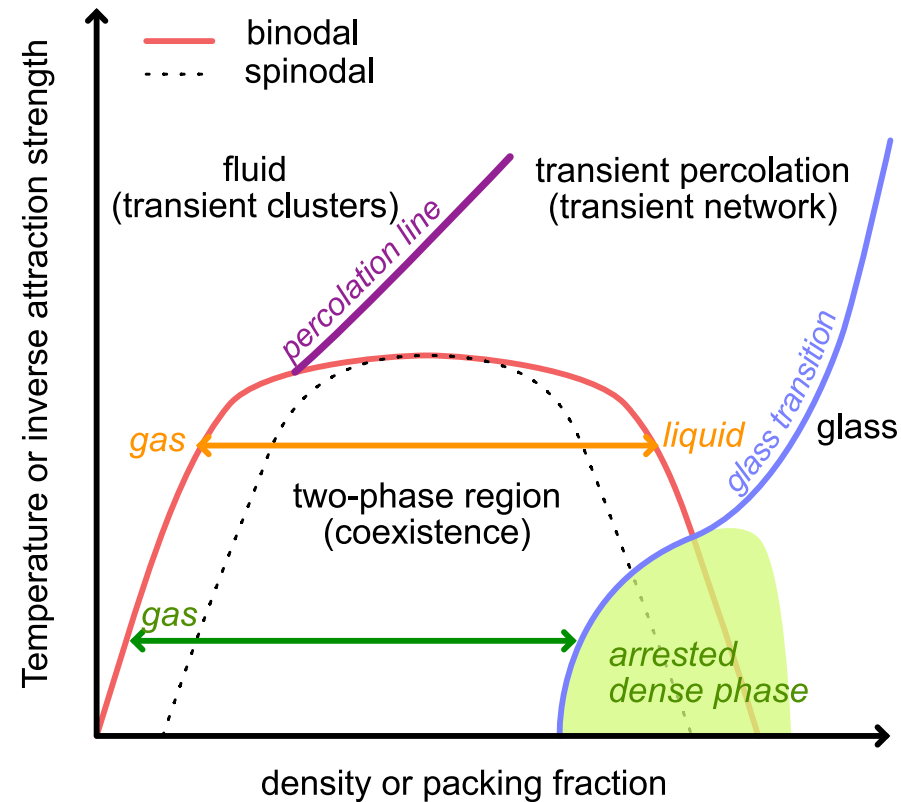
Scenario:

1. Rapid quench into unstable region
2. Phase separation begins (spinodal)
3. Dense regions arrest (glass/jamming)
4. Bicontinuous frozen structure

The picture suggest the progressive **freezing of coarsening upon spinodal decomposition.**



Phase Diagram: Arrested Spinodal



Arrested spinodal scenario, see Zaccarelli et al 2007

Key lines:: - Binodal (red): liquid-gase coexistence at equilibrium - Spinodal (dashed): limit of stability of the metastable phases. Instability (i.e. spinodal decomposition) starts **inside** - Percolation (purple): At some density, the clusters percolate (span the system) - Glass transition (blue): Dense phase becomes dynamically arrested (non-ergodic)

Gel path: Rapid cooling → arrested dense phase + vapor